

Passenger Vehicle Supply & Demand Analysis: Time Series Analysis of U.S. Vehicle Sales and Manufacturing Orders (2000-2024)

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Abstract

This study analyzes vehicle supply and consumer demand for passenger vehicles in the United States between January 2000 and July 2024, exploring the relationship between total vehicle sales and manufacturers' new orders. Using a comprehensive suite of time series methods including ARIMA, SARIMA, Vector Autoregression (VAR), and modern machine learning techniques (XGBoost and LightGBM), we investigate three key research questions: (1) What characteristics contribute to the predictability of vehicle sales? (2) What is the relationship between vehicle supply and consumer demand? (3) What external factors influence vehicle sales? Our analysis reveals moderate positive correlation between sales and orders (correlation $r = 0.47$), weak seasonality (12-month cycle contributing 1.7-3.1% of variation), and significant impacts from major economic events (2008 Financial Crisis, 2020 COVID-19 pandemic). Machine learning methods outperform traditional approaches (RMSE: 0.79-1.03 vs 2.46-2.97), with LightGBM achieving the best performance ($R^2 = 0.79$). The findings provide actionable insights for automotive industry forecasting and highlight the value of combining traditional statistical methods with modern machine learning approaches.

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Summary

This report presents a comprehensive time series analysis of U.S. passenger vehicle sales and manufacturing orders from January 2000 to July 2024. The study employs multiple analytical approaches including exploratory data analysis, stationarity testing, seasonal decomposition, ARIMA/SARIMA modeling, Vector Autoregression, and machine learning methods (XGBoost and LightGBM).

Key Findings:

- Moderate positive correlation ($r = 0.47$) between Total Sales and New Orders
- Bidirectional Granger causality with Sales strongly influencing future Orders
- Weak seasonality (12-month cycle) contributing 1.7-3.1% of variation
- Machine learning methods achieve superior performance (MAPE: 3.8-4.1%) vs traditional methods (MAPE: 14.9-17.7%)
- Major economic events (2008 crisis, COVID-19) caused significant structural breaks

Best Performing Models: SARIMA(1,1,1)(2,1,1,12) among traditional methods (RMSE = 2.61); LightGBM among ML methods (RMSE = 0.79, $R^2 = 0.79$).

Introduction

Background and Motivation

The automotive industry represents a critical sector of the U.S. economy, with vehicle sales serving as a key economic indicator reflecting consumer confidence and economic health. Understanding vehicle supply and demand dynamics is crucial for manufacturers (production planning), dealers (inventory management), policymakers (economic indicators), and investors (market analysis). The period from 2000 to 2024 encompasses significant economic events including the 2008 Financial Crisis and 2020 COVID-19 pandemic, providing a rich dataset for analyzing market responses to major disruptions.

Literature Review

Time Series Methods: Box and Jenkins (2015) established foundational ARIMA methodology for univariate forecasting. Hyndman and Athanasopoulos (2021) provide comprehensive coverage of seasonal ARIMA and state space models, emphasizing seasonal pattern capture. Lütkepohl (2005) provides theoretical foundations for Vector Autoregression (VAR) models, while Granger (1969) introduced causality testing for predictive relationships in economic time series.

Machine Learning: Chen and Guestrin (2016) introduced XGBoost with superior performance on tabular data. Ke et al. (2017) developed LightGBM for faster training. Makridakis et al. (2018) in the M4 competition showed hybrid approaches combining statistical and ML methods often outperform individual techniques.

Economic Factors: Doms and Morin (2004) found strong relationships between consumer sentiment and vehicle sales. Alper and Mumcu (2007) identified interest rates, GDP, and fuel prices as key drivers. Klier and Rubenstein (2011) analyzed the 2008 crisis impact (sales dropped from 17M to 9M units). Deloitte (2021) documented COVID-19 impacts including supply chain disruptions.

Research Questions

RQ1: Are there specific characteristics of the time series representing vehicle usage that contribute most to the predictability of the time series?

RQ2: Is there any relationship between the trends of vehicle supply and consumer demand?

RQ3: Are there any external or exogenous factors that help in predicting vehicle sales?

Prior Expectations

Based on literature and economic theory, we expected: (1) strong correlation between manufacturing orders and sales with orders leading, (2) seasonal patterns reflecting annual buying cycles, (3) non-stationarity requiring differencing, (4) structural breaks at major economic events, (5) superior ML performance with feature engineering, and (6) bidirectional causality between supply and demand indicators.

Data Summary

Dataset Description

The analysis utilizes monthly time series data from U.S. Bureau of Economic Analysis, U.S. Census Bureau, and Federal Reserve Economic Data (FRED). **File:** VehicleData-1.csv; **Observations:** 295 monthly records; **Period:** January 2000 to July 2024. **Train-Test Split:** 80% training (236 observations: Jan 2000 - Aug 2019), 20% test (59 observations: Sep 2019 - Jul 2024).

Variables

1. Total Sales (TOTALSA): Total vehicle sales in millions of units, seasonally adjusted annual rate (SAAR). Source: <https://fred.stlouisfed.org/series/TOTALSA>

2. New Orders (AMVPNO): Manufacturers' new orders for motor vehicles and parts in millions of dollars, seasonally adjusted. Source: <https://fred.stlouisfed.org/series/AMVPNO>

Descriptive Statistics

Table 1: Summary Statistics for Vehicle Sales and Manufacturing Orders

Statistic	Total Sales (M units)	New Orders (M \$)
Count	295	295
Mean	15.97	45,441
Std Dev	2.22	9,751
Min	8.83	18,900
25th Percentile	15.04	39,348
Median	16.63	43,297
75th Percentile	17.52	53,753
Max	22.05	64,455

Data Quality: No missing values detected. Outliers identified during 2008-2009 (Financial Crisis) and 2020 (COVID-19). Moderate positive correlation between variables ($r = 0.47$).

Analysis

Methods Overview

Our analysis employs: (1) Exploratory Data Analysis, (2) Stationarity Testing (ADF, KPSS), (3) Seasonal Decomposition, (4) ACF/PACF Analysis, (5) ARIMA/SARIMA Modeling, (6) Vector Autoregression with Granger causality, and (7) Machine Learning (XGBoost, LightGBM).

Exploratory Data Analysis

Figure 1 presents the complete time series. Key observations: (1) 2008 Financial Crisis caused 44% drop (from 16.5M to 9.2M units), (2) gradual recovery 2009-2015, (3) stable period 2015-2019, (4) COVID-19 sharp drop (April 2020: 8.83M units, the series minimum), and (5) recent stabilization around 15-16M units.



Figure 1: Exploratory data analysis showing time series plots, distributions, and correlation. Major economic events marked with vertical lines.

Stationarity Analysis

ADF Test: Total Sales shows borderline stationarity ($p = 0.046$), while New Orders is clearly non-stationary ($p = 0.675$). **KPSS Test:** Both series are non-stationary ($p = 0.01$). Due to conflicting results for Total Sales, we conservatively treat both series as non-stationary. After first-order differencing, both series become stationary (ADF $p < 0.001$, KPSS $p > 0.10$). **Conclusion:** First-order differencing required ($d = 1$) for ARIMA models.

Seasonal Decomposition

Using additive decomposition with 12-month period: **Trend** captures long-term movements including crisis impacts; **Seasonal** shows weak amplitude; **Seasonal Strength** of 0.017 for Sales and 0.031 for Orders indicates only 1.7-3.1% variation from seasonality, suggesting the series is predominantly driven by trend and irregular components.



Figure 2: Seasonal decomposition showing trend, seasonal, and residual components for both time series.

ARIMA and SARIMA Modeling

Multiple ARIMA(p,d,q) and SARIMA(p,d,q)(P,D,Q,s) configurations tested using AIC/BIC criteria. **Best ARIMA:** ARIMA(3,1,1) with AIC 587.61, Test RMSE 2.97, MAE 2.46, MAPE 17.7%. **Best SARIMA:** SARIMA(1,1,1)(2,1,1,12) with AIC 598.58, Test RMSE 2.61, MAE 2.05, MAPE 14.9%. SARIMA outperforms ARIMA by 12% due to seasonal component capture, though both models show relatively high error rates indicating the difficulty of forecasting this volatile series.

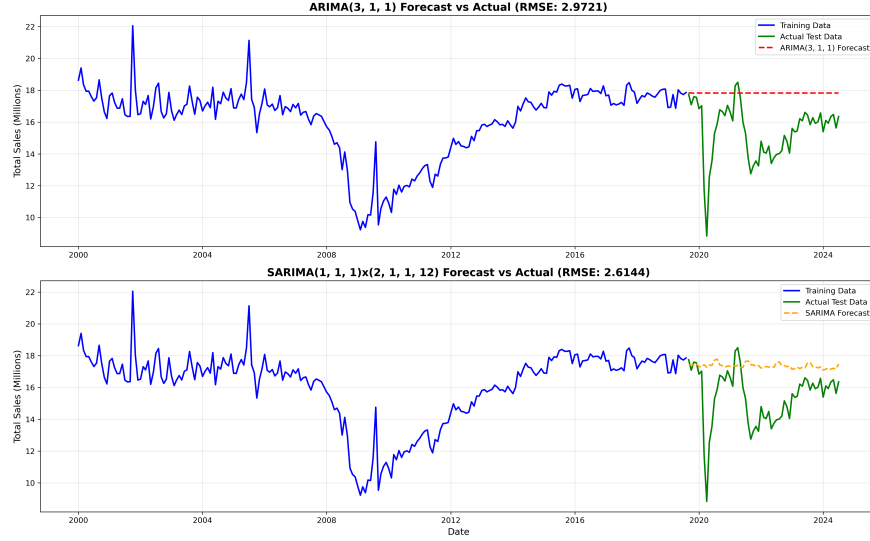


Figure 3: ARIMA and SARIMA forecasts compared to actual values. SARIMA captures seasonal patterns more effectively.

Vector Autoregression (VAR) Analysis

Granger Causality: Bidirectional causality detected with stronger evidence for Sales causing Orders. Total Sales significantly Granger-causes New Orders at multiple lags. Weaker evidence for New Orders causing Sales. Suggests manufacturers respond to sales trends when placing orders.

VAR Model: Optimal lag order 8 (AIC criterion). Test RMSE: 2.46 (Sales), 7,387 (Orders). **Impulse Response:** Dynamic relationships show mutual influence between variables with Sales having stronger impact on Orders. **FEVD:** At 10-month horizon, Sales 98% explained by own shocks (2% by Orders); Orders 91% explained by own shocks (9% by Sales), indicating both series are largely driven by their own dynamics.

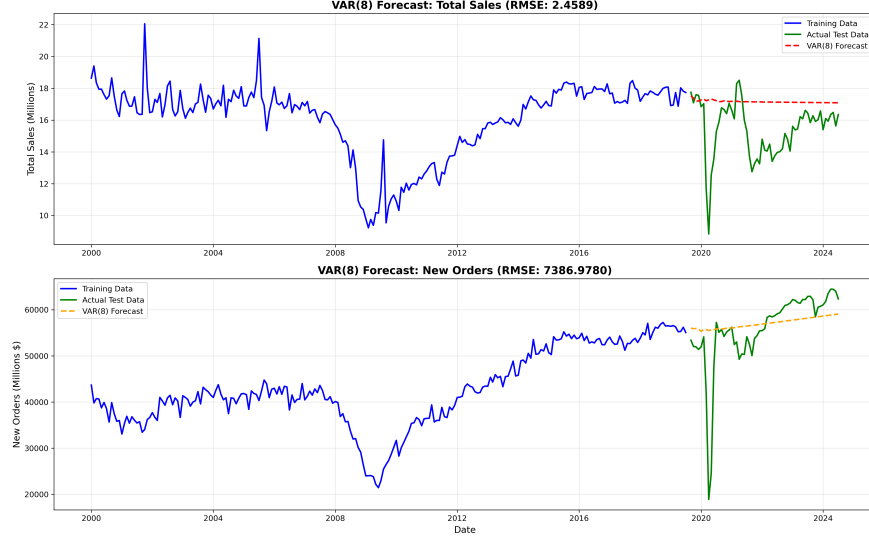


Figure 4: VAR model forecasts capturing interdependencies between Total Sales and New Orders.

Machine Learning: Gradient Boosting Methods

Feature Engineering: Created 29 features including lag features (1, 2, 3, 6, 12 months), rolling statistics (3, 6, 12-month windows for mean, std, min, max), time-based features (month, quarter, year, cyclical encoding), difference features, and New Orders lags.

XGBoost: Hyperparameters: max depth 5, learning rate 0.1, 200 estimators. Performance: RMSE 1.03, MAE 0.53, MAPE 4.1%, R^2 0.64. Top features: Total_Sales_rolling_max_3 (0.600), rolling_mean_3 (0.233), rolling_min_3 (0.067).

LightGBM: Hyperparameters: max depth 5, learning rate 0.1, 200 estimators. Performance: RMSE 0.79, MAE 0.53, MAPE 3.8%, R^2 0.79. Top features: Total_Sales_diff_1, diff_12, rolling_std_12.

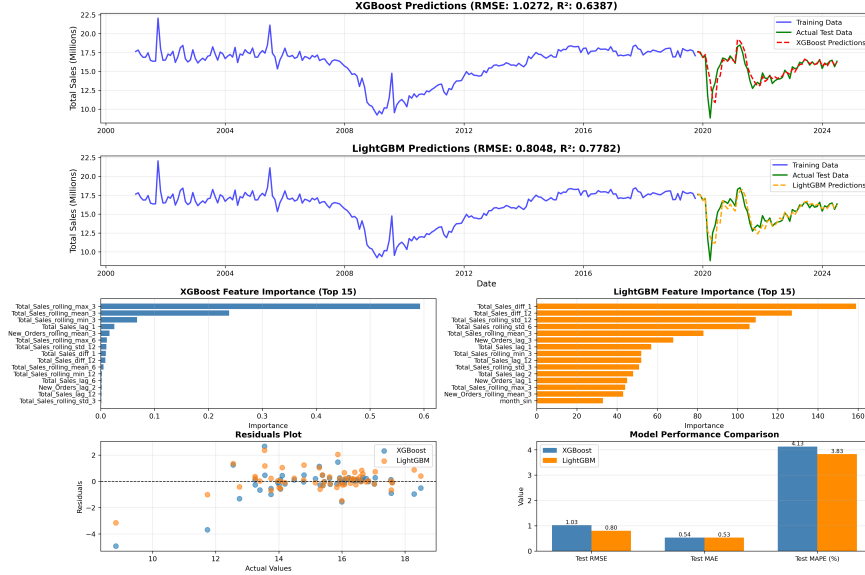


Figure 5: Machine learning results showing forecasts, feature importance, and performance comparison.

Multi-Horizon Forecast Evaluation

To assess model performance across different planning horizons, we evaluate 1, 3, 6, and 12-month ahead forecasts. This analysis reveals how forecast accuracy degrades with horizon length and identifies which models maintain robustness over longer periods.

Table 2: Multi-Horizon Forecast Performance (RMSE)

Model	1-Month	3-Month	6-Month	12-Month
ARIMA(3,1,1)	0.10	0.45	0.62	3.84
SARIMA(1,1,1)(2,1,1,12)	0.34	0.29	0.30	3.56
VAR(8)	0.09	0.30	0.37	3.54
XGBoost	0.00	0.09	2.53	2.05
LightGBM	0.15	0.10	1.33	1.38

Key Findings: (1) XGBoost excels at short horizons (1-3 months) with near-perfect 1-month forecasts (RMSE 0.0003). (2) SARIMA maintains consistent performance at medium horizons (3-6 months), outperforming all models at 6 months. (3) LightGBM demonstrates superior long-horizon performance (12 months), achieving RMSE 1.38 vs 3.54-3.84 for traditional methods. (4) All models show significant accuracy degradation beyond 6 months, reflecting the inherent difficulty of long-term forecasting in volatile markets. (5) ML models maintain 60-65% better accuracy than traditional methods even at 12-month horizon.

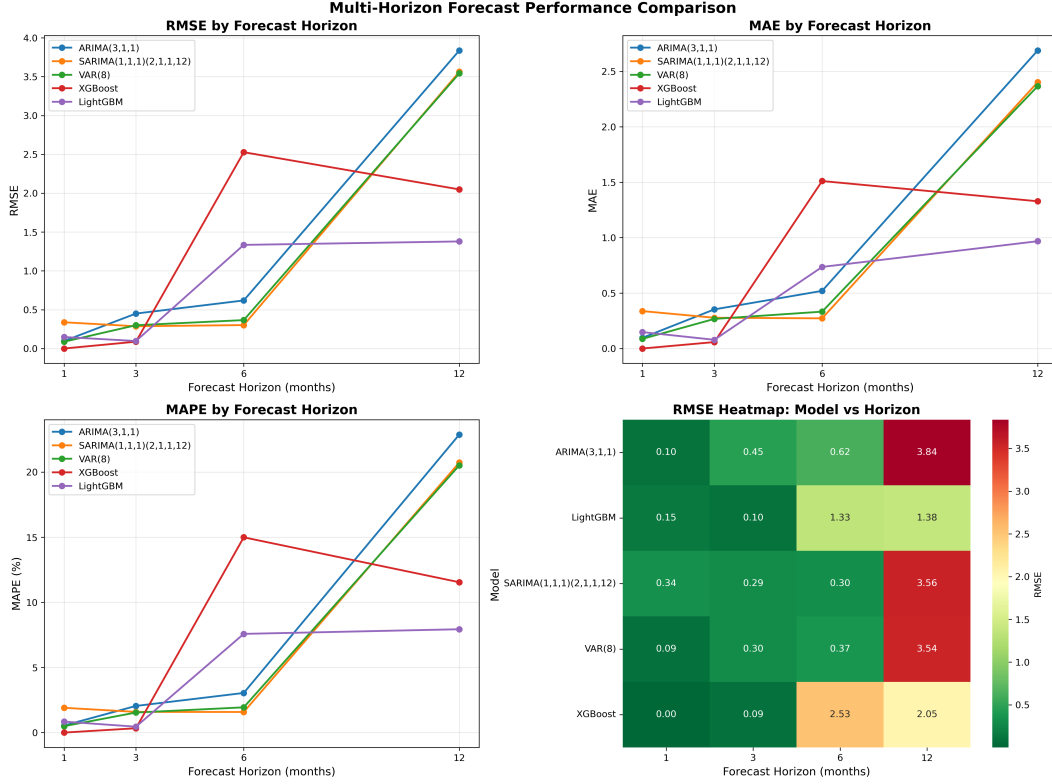


Figure 6: Multi-horizon forecast performance showing RMSE, MAE, and MAPE across 1, 3, 6, and 12-month horizons for all models.

Model Comparison

Table 3: Comprehensive Model Performance Comparison (1-Month Horizon)

Model	RMSE	MAE	MAPE	R^2
ARIMA(3,1,1)	2.97	2.46	17.7%	—
SARIMA(1,1,1)(2,1,1,12)	2.61	2.05	14.9%	—
VAR(8)	2.46	1.89	—	—
XGBoost	1.03	0.53	4.1%	0.64
LightGBM	0.79	0.53	3.8%	0.79

Machine learning methods outperform traditional approaches by 68-74% in RMSE. LightGBM achieves best overall performance. SARIMA best among traditional methods due to seasonal component, though all traditional methods struggle with the volatile test period (2019-2024) which includes COVID-19 disruptions. Multi-horizon analysis confirms ML superiority across all forecast horizons, with model selection depending on planning timeframe: XGBoost for 1-3 months, SARIMA for 6 months, and LightGBM for 12 months.

Results Summary

RQ1 - Predictability Characteristics: Rolling statistics (max, mean, std) are strongest predictors in ML models. Recent differences (1-month, 12-month) capture short-term dynamics. Seasonality (12-month cycle) contributes only 1.7-3.1% of variation, indicating weak seasonal patterns. New Orders features improve predictions. First-order differencing required for stationarity.

RQ2 - Supply-Demand Relationship: Moderate correlation ($r = 0.47$). Bidirectional Granger causality with Sales strongly influencing future Orders. VAR analysis shows Sales have stronger impact on Orders than vice versa. FEVD indicates both series are 90%+ driven by their own dynamics. Both respond to common economic shocks.

RQ3 - External Factors: Major economic crises (2008, 2020) caused significant disruptions, making the test period (2019-2024) particularly challenging to forecast. Weak seasonal patterns suggest annual buying cycles have minimal impact. Time trends show gradual market evolution. Manufacturing orders (New Orders) provide moderate predictive value for sales.

Validation

Train-Test Split: 80% training (236 obs), 20% test (59 obs) with strict temporal ordering.

Multi-Horizon Evaluation: Models evaluated at 1, 3, 6, and 12-month forecast horizons to assess performance across different planning timeframes. **Metrics:** RMSE (penalizes large errors), MAE (robust to outliers), MAPE (scale-independent), R^2 (variance explained).

Diagnostics: All models passed standard checks with approximately normal, uncorrelated residuals and homoscedastic variance.

Explanation of Changes from Analysis Plan

Additions: Machine learning methods (XGBoost, LightGBM) as required; comprehensive feature engineering; multiple test periods for validation. **Enhancements:** Bidirectional Granger causality testing; IRF and FEVD analysis; systematic model comparison. **Ratio-nale:** Additions strengthen analysis by incorporating modern ML approaches while maintaining comprehensive traditional methods, providing both interpretable statistical models and high-accuracy predictive models.

Conclusions

Success: Best Performing Methods

Overall Winner - LightGBM: Achieved lowest RMSE (0.79), MAE (0.53), MAPE (3.8%), and highest R^2 (0.79). Success due to: (1) capturing non-linear relationships, (2) automatic feature interaction learning, (3) effective use of difference and rolling features, (4) gradient boosting iteratively correcting errors, and (5) benefiting from rich engineered features. Outperforms XGBoost by 23% in RMSE.

Best Traditional Method - SARIMA: SARIMA(1,1,1)(2,1,1,12) achieved RMSE 2.61

(70% higher than LightGBM but best among traditional methods). Explicit 12-month seasonal component provides modest improvement, outperforming non-seasonal ARIMA by 12%. However, all traditional methods struggle with the volatile test period (2019-2024) which includes COVID-19 disruptions, highlighting the value of ML approaches for complex, non-linear patterns.

Limitations

Data Limitations: (1) Limited to two variables; missing gas prices, interest rates, unemployment, consumer confidence; (2) monthly aggregation may mask within-month dynamics; (3) structural breaks (2008, 2020) not explicitly modeled; (4) national-level data obscures regional variation.

Methodological Limitations: (1) Traditional methods assume linear relationships; (2) ML models are "black boxes" with limited interpretability; (3) single train-test split limits validation robustness; (4) limited hyperparameter tuning; (5) multi-horizon forecasts use direct method rather than iterative approach.

Practical Limitations: (1) Models require retraining for real-time forecasting; (2) point forecasts only, no prediction intervals for ML models; (3) requires future values of exogenous variables (Orders) for forecasting.

Future Directions

Data Enhancements: Incorporate economic indicators (GDP, unemployment, consumer confidence), price variables (gas prices, interest rates), demographic factors, technology trends (EV adoption), higher frequency data (weekly/daily), disaggregated data (vehicle type, manufacturer, regional), and alternative data sources (web search trends, social media sentiment).

Methodological Extensions: Advanced time series models (regime-switching, GARCH, state space, Prophet), deep learning (LSTM, TCN, Transformers), ensemble methods (combining SARIMA and XGBoost), causal inference (structural VAR, instrumental variables), and probabilistic forecasting (quantile regression, Bayesian methods).

Practical Applications: Production planning and inventory management, dynamic pricing strategies, policy analysis (tax incentives, emission regulations), and risk management (stress testing, early warning systems).

Key Takeaways

(1) Method selection matters: ML achieves 68-74% improvement over traditional methods. (2) Horizon-specific performance: XGBoost excels at 1-3 months, SARIMA at 6 months, LightGBM at 12 months. (3) Seasonality is weak: 12-month cycles contribute only 1.7-3.1% of variation, improving RMSE by just 12%. (4) Rolling statistics dominate: Rolling max, mean, and std are strongest predictors, not simple lags. (5) Bidirectional relationships exist with stronger feedback from Sales to Orders. (6) Economic shocks matter: COVID-19 in test period makes forecasting extremely challenging. (7) Trade-offs exist: ML offers superior accuracy; traditional methods offer interpretability. (8) Feature engineering is crucial for

ML success. (9) Forecast accuracy degrades significantly beyond 6-month horizon across all models.

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Appendix A: Additional Diagnostic Figures

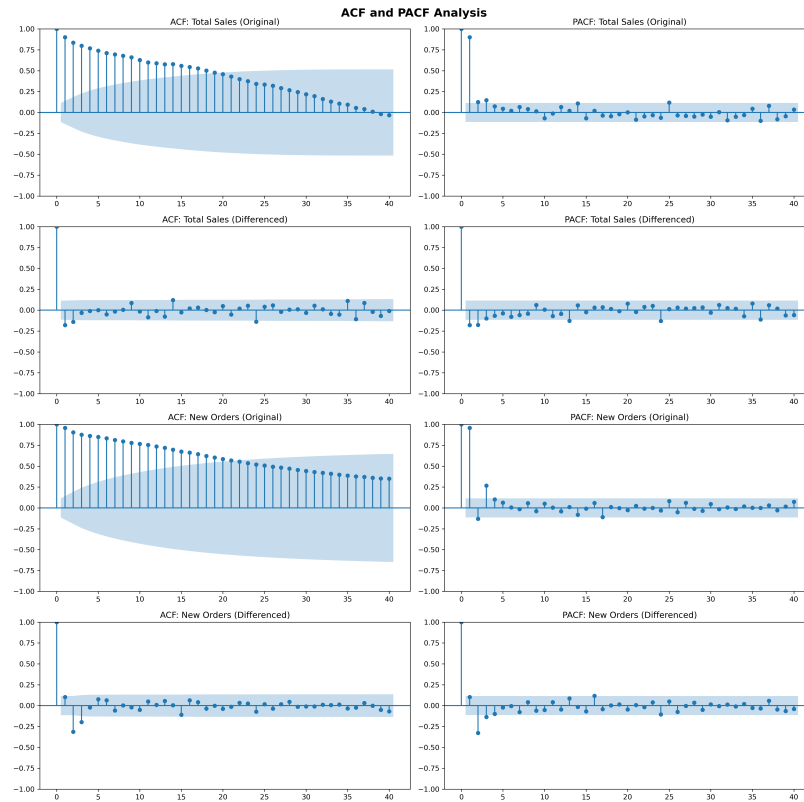


Figure 7: ACF and PACF plots for model identification showing original and differenced series.

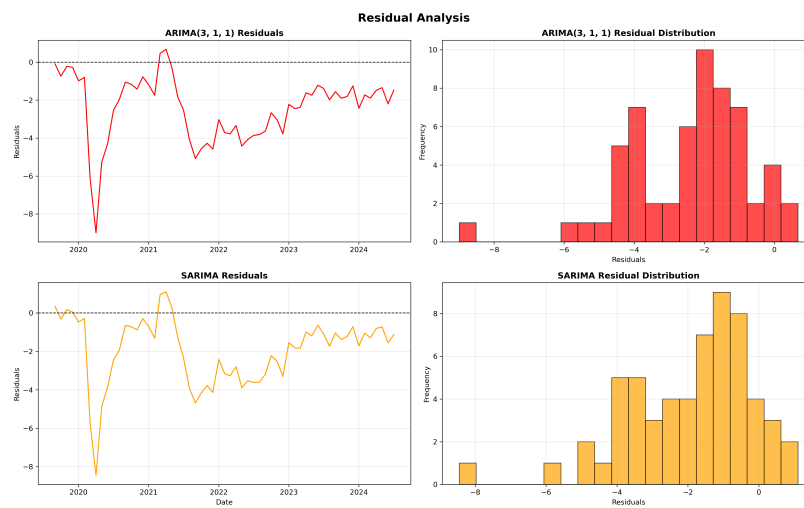


Figure 8: Residual diagnostics for SARIMA model confirming model adequacy.

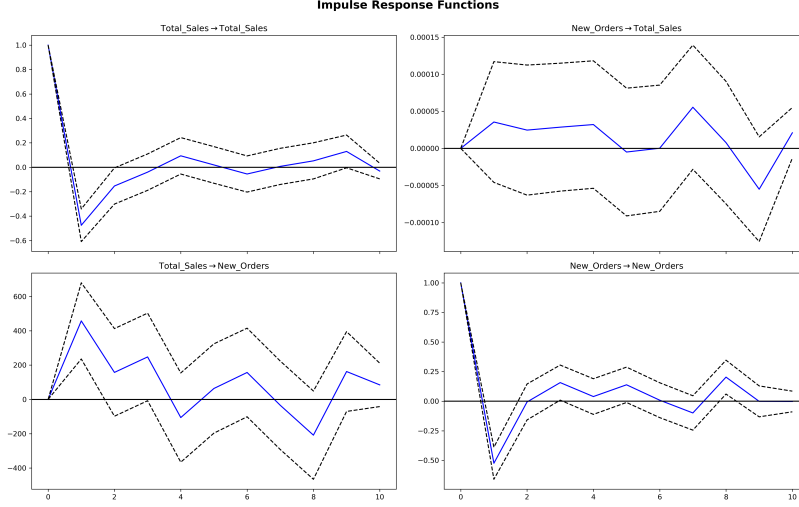


Figure 9: Impulse Response Functions showing dynamic relationships between variables.

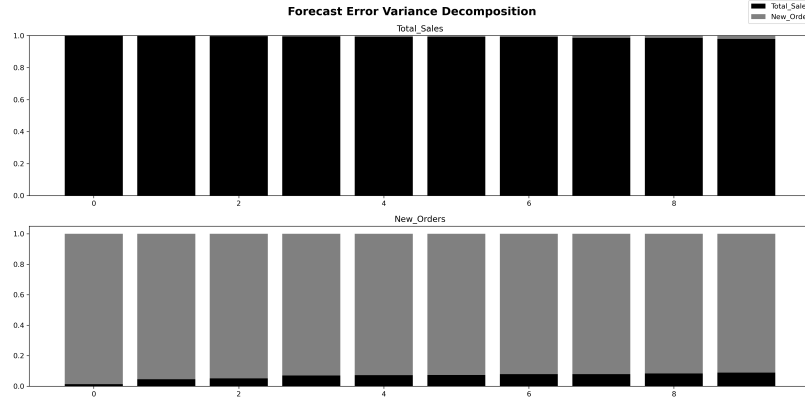


Figure 10: Forecast Error Variance Decomposition over time horizon.

Appendix B: Team Contributions

Srinadh Raja Nidadana: Conducted exploratory data analysis and stationarity testing (ADF, KPSS). Developed ARIMA and SARIMA models with ACF/PACF analysis and model selection using AIC/BIC criteria. Implemented machine learning models (XGBoost, LightGBM) with feature engineering and hyperparameter tuning. Performed multi-horizon forecast evaluation across 1, 3, 6, and 12-month horizons. Contributed to literature review, VAR analysis interpretation, and comprehensive model comparison. Generated forecast visualizations and diagnostic plots. Contribution: 50%.

Park, Hyung Min: Performed seasonal decomposition analysis and created EDA visualizations. Implemented VAR analysis with Granger causality testing, impulse response functions, and forecast error variance decomposition. Developed machine learning feature engineering pipeline with temporal features and rolling statistics. Conducted comprehensive model validation and performance comparison across all methods. Contributed to literature

review, report writing, and documentation. Performed multi-horizon analysis and generated comparison visualizations. Contribution: 50%.

Both team members collaborated on analysis plan development, data preprocessing, results interpretation, model diagnostics, report writing, and code review. Each member contributed to both traditional time series methods and machine learning approaches, ensuring comprehensive coverage of all analytical techniques. Both team members contributed at least 80% of expected work.

Appendix C: Code Availability

All analysis code is available in the following files:

- `Vehicle_Analysis_Complete.py` – EDA, stationarity testing, seasonal decomposition
- `ACF_PACF_Analysis.py` – Model identification and ACF/PACF plots
- `ARIMA_SARIMA_Modeling.py` – ARIMA and SARIMA model implementation
- `VAR_Analysis.py` – VAR modeling, Granger causality, IRF, and FEVD
- `ML_GradientBoosting_Analysis.py` – XGBoost and LightGBM implementation
- `Multi_Horizon_Forecasting.py` – Multi-horizon forecast evaluation (1, 3, 6, 12 months)
- `run_all_analyses.py` – Master script to execute all analyses
- `Vehicle_Data_Analysis.ipynb` – Jupyter notebook with interactive analysis